University of California, Berkeley
Office of Environment, Health & Safety - Radiation Safety

Radiation Use Authorization

RUA Number: **3271** -- **Radioactive Materials** -- **Active** HGV: 665.12110 Class: II Survey Frequency: **Semi-Annual**

RUA Holder: **Donald J. ORLANDO** - phylabs@berkeley.edu - +1 510 642-5328 Department: **Physics** Amendment Number: 1 Revision Number: 13

Expiration: **07/31/2016**

Location(s): LeConte Hall: 286

Lab Contact(s): Donald J. ORLANDO - +1 510 642-5328 - phylabs@berkeley.edu ;

Personnel: Donald J. ORLANDO *;

** indicates personnel who are specifically required to use dosimetry.*

Radioradionuclide Limits

Radionuclide	Chemical Form(s)	Physical Form	mCi/Expt.	mCi/Vial	mCi Poss	mCi/Year
Na22	Two sources; 1 in epoxy matrix, 1 disc	Solid	0.100	0.100	0.200	0.200
Ra226	sealed source	Solid	0.010	0.010	0.010	0.010
Fe55	Three sources	Solid	5.000	5.000	11.534	11.534
Cs137	sealed sources	Solid, Liquid, Gas	2.500	2.500	4.000	4.000
Sr90	sealed source	Solid, Liquid, Gas	20.000	20.000	20.000	20.000
H3	source	Solid	0.111	0.111	0.111	0.111
Ag110m	foil in brass housing	Solid	0.111	0.111	0.111	0.111
Mn54	Four sources	Solid	0.200	0.200	0.400	0.400
Co60	Two sources in epoxy matrix	Solid	0.100	0.100	0.200	0.200
Am241	Five unsealed sources	Solid	0.260	0.260	0.616	0.616
Co57	Four unsealed source	Solid	16.000	16.000	40.000	40.000

Experiment Description

1) Use and storage of solid and sealed radioactive sources in the study of various laws of physics for Physics 111 Lab.

2) Perform source insertion, decontamination, repair, and/or maintenance of beta chamber

Precautions Required

The following protection items are required: Lab Coat, Disposable Gloves, Eye Protection Equipment, and if marked, the following:

☒ TLD Ring ☐ Body Badge ☐ Thyroid Scan ☐ Urinalysis ☐ Neutron Badge ☒ G-M Meter ☐ Scint Probe
☐ Neuton Meter ☐ Alpha Probe ☐ Ion Chamber ☐ IC ☐ E/D Exception Granted ☒ () ACUC Protocol

1. Compliance: With provisions and requirements of UC Berkeley\\'s RSM and policy on food & beverages in laboratories (http://campuspol.chance.berkeley.edu/policies/foodinlabs.pdf).
2. Notifications: Immediately notify RSO: of any missing or leaking sources; suspected personnel radiation exposure; contamination found on the equipment, personnel, or in the immediate area; lost or misplaced sources; exceeding RUA limits, or sources transferred out of authorized locations.
3. Training: RUA Holder, Lab Contact, and any individual working on equipment housing sources are required to complete unsealed RAM user training. Classroom students are required to complete sealed source training and additional training required by Physics Department prior to starting work activities. RUA Holder is required to maintain a list of students attending classroom work and provide appropriate level of training as approved by EH&S RST trainer.
4. Dosimetry: Rings required when handling sources > 1 mCi. Handle, store, and return dosimeters at specified frequencies. Notify EH&S of lost dosimeters. Dosimeters may be issued for classroom work at RUA Holders request and follow both EH&S and Physic's requirements. Contact EH&S for assistance with classroom dosimetry. Classroom dosimetry is required to be returned at the end of the class period.
5. Inventory: Keep radioactive sources in secured, properly shielded, and labeled locations to prevent unauthorized removal. Do not leave radioactive sources unattended unless in a posted experiment. Ensure that all users maintain the radioactive materials inventory in RSIS and make timely updates of use, disposal, and/or transfer of these materials. Make sealed sources available for semi-annual leak tests and perform semi-annual inventory of all sources as required.
6. Posting/Storage: Label all radioactive material work/storage locations, with the trefoil and "Caution-Radioactive Materials". Label outside of containers with isotope, activity, and date. Post equipment with a radioactive material sign if sources are left out for an extended amount of time. Label exempt quantities of material with a trefoil.
7. Handling/Contamination: Performed work in designated areas. Wear disposable gloves when handling sources. Maintain routine dose rates < 5 mR/hr at 30 cm from the source. Contact EH&S if dose rates exceed 5 mR/hr at 30cm. Contact EH&S prior to replacing

source in chamber, accessing, or performing maintenance on beta chamber for controls.
8. Transportation: All rad material shall be received and transferred through HMF.
9. Waste: Contact EH&S regarding disposal of sources. All waste shall be disposed of through HMF. Disposable gloves will be discarded as contaminated waste. Segregate and handle waste in compliance with the Campus Radioactive Waste Guidelines. No sink disposal is authorized.
10. Surveys: Have available a portable G-M survey meter while working with sources. Survey personnel (hands, shoes, lab coat) and work areas (floor, bench, equipment, etc.) for loose contamination during and immediately after working with sources. No routine self-surveys are required.

Certification

All work will be performed with the precautions specified.

Signed RUA on file at RS

RUA Holder — Date

Signed RUA on file at RS

Department Chair — Date

Radiation use is approved with the required precautions.

Signed RUA on file at RS

Radiation Safety Officer — Date

Signed RUA on file at RS

Radiation Safety Committee — Date

Rev Date 07/28/2015 *FOR ASSISTANCE, CALL EH&S - RS (359 U-HALL MC1150) AT 2-3073 or FAX 3-9495* Last Update 07/28/2015